

PART-1 : MATHEMATICS

SECTION-I (i)

1) The number of possible value of ' α ' for which the expression $y = \frac{\alpha x^2 + 7x - 2}{\alpha + 7x - 2x^2}$ has atleast one common linear factor in numerator and denominator, is -

- (A) 0
- (B) 1
- (C) 2
- (D) 3

2) Let ΔABC has perimeter 20 units. Then $\lim_{n \rightarrow \infty} (a^n + (b+c)^n)^{\frac{1}{n}} + (b^n + (c+a)^n)^{\frac{1}{n}} + (c^n + (a+b)^n)^{\frac{1}{n}}$ is (with usual notations)

- (A) 20
- (B) 40
- (C) 80
- (D) 100

3) $\lim_{x \rightarrow 0} \left(2 + \log_{\sec(\frac{x}{2})} \cos\left(\frac{x}{3}\right) \right)^3$ is equal to

- (A) $\left(\frac{146}{81}\right)^3$
- (B) $\left(\frac{70}{27}\right)^3$
- (C) $\left(\frac{178}{81}\right)^3$
- (D) $\left(\frac{34}{9}\right)^3$

4) Let $f(t) = |t| + |t - 1| \forall t \in \mathbb{R}$ and $g(x) = \begin{cases} \max.(f(t)), & x - 1 \leq t \leq x, \quad 0 \leq x \leq 1 \\ 3 - x, & 1 < x \leq 2 \end{cases}$. Find number of points where $g(x)$ is non-derivable in $[0, 2]$

- (A) 0
- (B) 1
- (C) -1

(D) 3

SECTION-I (ii)

1) Suppose that $f(a) > 0$ and that f is differentiable at $x = a$. Which of the following statements is/are TRUE.

(A) $\lim_{n \rightarrow \infty} \left(\frac{f\left(a + \frac{1}{n}\right)}{f(a)} \right)^{\frac{1}{n}} = 1$

(B) $\lim_{n \rightarrow \infty} \left(\frac{f\left(a + \frac{1}{n}\right)}{f(a)} \right)^{\frac{1}{n}} = e^{\frac{f'(a)}{f(a)}}$

(C) $\lim_{x \rightarrow a^+} \left(\frac{f(x)}{f(a)} \right)^{\frac{1}{\ln x - \ln a}} = e^{\frac{af'(a)}{f(a)}}; a > 0$

(D) $\lim_{x \rightarrow a^-} \left(\frac{f(x)}{f(a)} \right)^{\frac{1}{\ln x - \ln a}} = e^{\frac{af'(a)}{f(a)}}; a > 0$

2) $\lim_{x \rightarrow 0} \frac{\sin(\sin x) - \sin x}{ax^5 + bx^3 + c} = -\frac{1}{12}$, then

(A) $a = 2, b = 0, c = 1$

(B) $a \in \mathbb{R}, b = 2, c = 0$

(C) $a \in \mathbb{R}, b - c = 2$

(D) $a \in \mathbb{R}, b + c = 2$

3) Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be defined by $f(x) = (x - 2)^2 \cos\left(\frac{\pi x}{4}\right) + (x - 2)|x - 2|$ and let $g : \mathbb{R} \rightarrow \mathbb{R}$ be an arbitrary function. Define the product function $h : \mathbb{R} \rightarrow \mathbb{R}$ by $h(x) = f(x)g(x)$. Which of the following statements is/are TRUE?

(A) If $\lim_{x \rightarrow 2} g(x)$ exists, then h is differentiable at $x = 2$.

(B) If g is bounded in an open interval containing $x = 2$, then h is differentiable at $x = 2$.

(C) If h is differentiable at $x = 2$, then g must be continuous at $x = 2$.

(D) If h is differentiable at $x = 2$, then $\lim_{x \rightarrow 2} g(x) = 0$.

SECTION-I (iii)

1) Match List-I with List-II and select the correct answer using the code given below the list.

List-I		List-II	
(P)	$\lim_{x \rightarrow \infty} \frac{1}{\pi} \tan^{-1}(x^2 - x^4)$ is equal to	(1)	1

(Q)	$\lim_{x \rightarrow \infty} \frac{e^{x \ln 2}}{e^x}$ is equal to	(2)	$-\frac{1}{2}$
(R)	If $y = f \circ f \circ f(x)$ is differentiable function where $f(0) = 0$ and $f'(0) = 2$, then value of $y'(0)$ is equal to	(3)	0
(S)	$\lim_{x \rightarrow 2^-} \frac{[x]}{\{x\}}$ is equal to (Here $[.]$ & $\{.\}$ denotes greatest integer & fractional function respectively)	(4)	8
		(5)	10

(A) $P \rightarrow 2; Q \rightarrow 3; R \rightarrow 4; S \rightarrow 1$

(B) $P \rightarrow 1; Q \rightarrow 2; R \rightarrow 3; S \rightarrow 4$

(C) $P \rightarrow 2; Q \rightarrow 4; R \rightarrow 3; S \rightarrow 1$

(D) $P \rightarrow 3; Q \rightarrow 1; R \rightarrow 4; S \rightarrow 5$

2) Match the functions in List I with their evaluated parameters in List II.

List-I		List-II	
(P)	$f(x) = \begin{cases} Ax^a \left(x - \frac{1}{x}\right), & x > 0 \\ e^x, & x \leq 0 \end{cases}$ Let $f(x)$ be continuous at $x = 0$. Then the value of $(A + a)$ is	(1)	0
(Q)	$f(x) = \begin{cases} \frac{\ln(1+ax) - \ln(1-bx)}{x}, & x < 0 \\ k, & x = 0 \\ \frac{\sqrt{1+2x} - \sqrt{1-2x}}{\sin x}, & x > 0 \end{cases}$ Let $f(x)$ be continuous at $x = 0$ with $a = 1$. Then the value of $(k - b)$ is	(2)	1
(R)	$f(x) = \begin{cases} \frac{1 - \cos(2x)}{x^2}, & x > 0 \\ c, & x = 0 \\ \frac{2x - \sin(2x)}{x^3} + \frac{2}{3}, & x < 0 \end{cases}$ Let $f(x)$ be continuous at $x = 0$. Then the value of c is	(3)	2
(S)	$f(x) = \begin{cases} \frac{\sqrt{1+\sqrt{1+kx^4}} - \sqrt{2}}{x^4}, & x \neq 0 \\ c, & x = 0 \end{cases}$ Let $f(x)$ be continuous at $x = 0$ with $c = \frac{1}{8\sqrt{2}}$. Then the value of $6k$ is	(4)	3
		(5)	7

(A) $P \rightarrow 2; Q \rightarrow 1; R \rightarrow 4; S \rightarrow 3$

(B) $P \rightarrow 1; Q \rightarrow 2; R \rightarrow 3; S \rightarrow 4$

(C) $P \rightarrow 4; Q \rightarrow 3; R \rightarrow 2; S \rightarrow 1$

(D) $P \rightarrow 3; Q \rightarrow 5; R \rightarrow 1; S \rightarrow 4$

3) Match the statements in List-I with their corresponding values in List-II.

List-I		List-II	
(P)	Let $f(x) = \begin{cases} ax^2 + b, & x \leq 1 \\ \frac{1}{ x }, & x > 1 \end{cases}$. If $f(x)$ is differentiable at $x = 1$, then the value of $(b - a)$ is	(1)	0
(Q)	Let $f(x) = \begin{cases} x^p \sin\left(\frac{1}{x}\right), & x \neq 0 \\ 0, & x = 0 \end{cases}$. The smallest integer p such that $f(x)$ is differentiable at $x = 0$ is	(2)	1
(R)	Let $f(x) = \max\{ x , x^2, x^3\}$ for $x \in [-2, 2]$. The number of points in $(-2, 2)$ where $f(x)$ is not differentiable is	(3)	2
(S)	Let $f(x) = \begin{cases} \cos(\pi x) + ax, & x < 0 \\ b(1 - x^2)^{1/3}, & x \geq 0 \end{cases}$. If $f(x)$ is differentiable at $x = 0$, then the value of $(a + b)$ is	(4)	3
		(5)	7

(A) $P \rightarrow 2; Q \rightarrow 2; R \rightarrow 1; S \rightarrow 4$

(B) $P \rightarrow 4; Q \rightarrow 4; R \rightarrow 3; S \rightarrow 1$

(C) $P \rightarrow 3; Q \rightarrow 3; R \rightarrow 4; S \rightarrow 2$

(D) $P \rightarrow 1; Q \rightarrow 5; R \rightarrow 2; S \rightarrow 3$

SECTION-II

1) Number of integral values of 'x' satisfying the equation

$$\left| 1 - \log_{\frac{1}{6}} x \right| + |\log_2 x| + 2 = \left| 3 - \log_{\frac{1}{6}} x - \log_2 x \right| \text{ is}$$

2) Let the solution set of the equation $\sqrt{\left[x + \left[\frac{x}{2}\right]\right]} + \left[\sqrt{\{x\}} + \left[\frac{x}{3}\right]\right] = 3$ is $[a, b)$. Find the sum $a + b$ (where $[.]$ and $\{.\}$ denote greatest integer and fractional part function respectively).

3) Let $f(x) = \frac{1 - \sin x}{(\pi - 2x)^2} \frac{\log \sin x}{\log(1 + \pi^2 - 4\pi x + 4x^2)} \left(x \neq \frac{\pi}{2}\right)$. The value of $f\left(\frac{\pi}{2}\right)$ so that the function is continuous at $x = \frac{\pi}{2}$ is $\frac{-1}{\lambda}$, then the value of λ is

4) If $y = \left(1 + \frac{1}{x}\right)^x$ then $\frac{\sqrt[2]{y_2(2) + \frac{1}{8}}}{\left(\log \frac{3}{2} - \frac{1}{3}\right)}$ is equal to (where $y_2(x)$ is $\frac{d^2 y}{dx^2}$)

5) Given $y = \frac{\sqrt{1+2x}\sqrt[4]{1+4x}\sqrt[6]{1+6x}\dots\sqrt[100]{1+100x}}{\sqrt[3]{1+3x}\sqrt[5]{1+5x}\sqrt[7]{1+7x}\dots\sqrt[101]{1+101x}}$, then the value of y' at $x = 0$, is

6) Let the polynomial equations $x^3 - 5x^2 + px + q = 0$ and $x^3 - 2x^2 + (p - 3)x + r = 0$ have two roots in common, and let the third roots of each equation be represented by α and β , respectively. If the

function $f(x) = \begin{cases} 2^{\frac{\sin(\alpha x)}{\ln(1+\beta x)}}, & -1 < x < 0 \\ a, & x = 0 \\ b \cdot \frac{\ln(e^{x^2} + \alpha\beta\sqrt{x})}{\tan\sqrt{x}}, & 0 < x < 1 \end{cases}$ is continuous at $x = 0$, then the value of $(a + b)$ is

PART-2 : PHYSICS

SECTION-I (i)

1) A transverse wave travelling along a stretched string is represented by $y(x,t) = 3 \cos(4\pi t - 2\pi x) + 4 \sin(4\pi t - 2\pi x)$ mm, where x is in metres and t is in seconds. At $t = 0$, a point P on the string has displacement $y_p = 4$ mm and is moving upward. Let Q be the nearest point to the left of P whose transverse velocity is zero at the same instant. The distance PQ and the transverse acceleration of Q at $t = 0$ are respectively :-

- (A) $\frac{1}{2\pi} \tan^{-1}\left(\frac{3}{4}\right)$ m, $-80\pi^2$ mm/s²
- (B) $\frac{1}{2\pi} \cos^{-1}\left(\frac{4}{5}\right)$ m, $+80\pi^2$ mm/s²
- (C) $\frac{1}{2\pi} \sin^{-1}\left(\frac{3}{5}\right)$ m, $+80\pi^2$ mm/s²
- (D) $\frac{1}{4}$ m, $+80\pi^2$ mm/s²

2) A point sound source S_1 produces a sound level of 80 dB at a point P. Another independent point source S_2 , placed at twice the distance from P as S_1 , has four times the acoustic power of S_1 . Both sources are switched on simultaneously. A third independent source S_3 is then switched on, and the total sound level at P increases by 3 dB. Assume all sources are isotropic and mutually incoherent, $10^{0.3} \approx 2$. The sound level produced by S_3 alone at P is :-

- (A) 77 dB
- (B) 80 dB
- (C) 83 dB
- (D) 86 dB

3) A standing electromagnetic wave in vacuum is formed by two coherent plane waves of equal amplitude E_0 , travelling along $\pm x$. The resultant electric field is $\vec{E}(x,t) = 2E_0 \sin(kx) \cos(\omega t)\hat{y}$, $\omega = ck$. At a point P in the region $0 < kx < \pi/2$, the instantaneous electric and magnetic energy densities

are equal at $t = \frac{\pi}{6\omega}$. At the same instant, the Poynting vector at P is directed along -x. The value of kx_p is :-

- (A) $\frac{\pi}{6}$
- (B) $\frac{\pi}{3}$
- (C) $\frac{\pi}{4}$
- (D) $\tan^{-1}\left(\frac{1}{\sqrt{2}}\right)$

4) A diffraction-limited surveillance telescope is used to distinguish two small LEDs mounted on a distant tower. The LEDs are separated horizontally by 2.0 cm. The telescope has a circular objective of diameter 5.0 cm and operates at wavelength 550 nm. Assume Rayleigh's criterion and neglect atmospheric effects. What is the maximum distance of the tower from the telescope for which the two LEDs can just be resolved?

- (A) 0.75 km
- (B) 1.49 km
- (C) 2.98 km
- (D) 4.47 km

SECTION-I (ii)

1) A sonometer wire of linear mass density μ is stretched under tension T. For a vibrating length L, it resonates with a tuning fork of frequency f in the fundamental mode. The same wire is then subjected to different changes. Assume the bridges are fixed and neglect end corrections. Which of the following statements are correct?

- (A) If the tension is increased to 9T and the vibrating length is increased to $3L/2$, the new fundamental frequency is 2f.
- (B) If the vibrating length is changed to $3L/2$ and the tension to 4T, the second harmonic of the wire has frequency $\frac{8f}{3}$.
- (C) If the wire is replaced by another wire of the same material but twice the radius, while T and L remain unchanged, the third harmonic of the new wire has frequency $\frac{3f}{2}$.
- (D) If the tension is changed to 4T and the vibrating length to 2L, the frequency of the second harmonic remains equal to 2f.

2) A non-absorbing thin film of refractive index $\mu = \frac{4}{3}$ and thickness $t = 450$ nm is deposited on a glass substrate of refractive index $n_g = \frac{3}{2}$. Light is incident normally from air. Assume only the two rays reflected from the upper and lower surfaces of the film interfere. Which of the following statements are correct ?

- (A) For incident wavelength 600 nm, the reflected rays interfere constructively.

- (B) For incident wavelength 480 nm, the reflected rays interfere destructively.
- (C) For 600 nm light, the minimum increase in film thickness required to change the reflected light from a maximum to a minimum is 112.5 nm.
- (D) If the substrate is replaced by a transparent material of refractive index 1.20, keeping $t = 450$ nm, then 600 nm light still gives a reflected maximum.

3) Two semi-infinite strings are joined at $x = 0$ and stretched under the same tension. Their linear mass densities satisfy $\mu_2 = 4\mu_1$. A transverse wave travelling along string 1 ($x < 0$) towards the junction is $y_i = 6 \cos(40t - 8x + \frac{\pi}{6})$ mm. The junction is massless and there is no dissipation. Which statements are correct ?

- (A) The reflected wave has amplitude 2 mm, undergoes a phase reversal, and is $y_r = 2 \cos(40t + 8x + \frac{\pi}{6})$ mm.
- (B) The transmitted wave has amplitude 4mm, half the incident wavelength, and is $y_t = 4 \cos(40t - 16x + \frac{\pi}{6})$ mm.
- (C) The junction executes SHM according to $y(0,t) = 4 \cos(40t + \frac{\pi}{6})$ mm, with maximum transverse speed 40 mm/s.
- (D) One-third of the incident average power is reflected.

SECTION-I (iii)

1) In four different two-beam interference arrangements, monochromatic light of wavelength $\lambda = 600$ nm is used. Small-angle and paraxial approximations may be used. Match each arrangement in List-I with the corresponding fringe width in List-II. Which of the following gives the correct matching ?

	List-I		List-II
(P)	Fresnel biprism : A narrow slit is at $a = 25$ cm from a biprism of refractive index 1.50. The refracting angle of each half is $A = 1.0 \times 10^{-3}$ rad. A screen is 75 cm beyond the biprism.	(1)	0.80 mm
(Q)	Lloyd's mirror : A point source is $h = 0.20$ mm above a plane mirror. A screen perpendicular to the mirror is placed 80 cm from the source along the mirror.	(2)	1.20 mm
(R)	Fresnel mirrors : Two plane mirrors inclined at a small angle $\theta = 0.75 \times 10^{-3}$ rad have a point source 30 cm from their intersection on the bisector. The screen is 90 cm from the mirror intersection. Take the separation of the coherent virtual images as $d \approx 2a\theta$.	(3)	1.60 mm

(S)	Billet split lens : A convex lens of focal length 20 cm has an axial point source 30 cm in front of it. The lens is split into two halves and their optical centres are displaced symmetrically by 0.10 mm on either side of the original axis. The screen is 80 cm beyond the plane containing the two images.	(4)	2.40 mm
		(5)	3.20 mm

(A) $P \rightarrow 4; Q \rightarrow 2; R \rightarrow 3; S \rightarrow 1$

(B) $P \rightarrow 3; Q \rightarrow 2; R \rightarrow 4; S \rightarrow 1$

(C) $P \rightarrow 4; Q \rightarrow 1; R \rightarrow 3; S \rightarrow 2$

(D) $P \rightarrow 2; Q \rightarrow 4; R \rightarrow 1; S \rightarrow 3$

2) In each case of List-I, a monochromatic plane electromagnetic wave propagates along +z in vacuum. The magnetic-field components are given. Match each case with the corresponding state/property of polarization in List-II. Which of the following gives the correct matching ?

	List-I		List-II
(P)	$B_x = -B_0 \sin \omega t, B_y = B_0 \cos \omega t$	(1)	Circularly polarized wave
(Q)	$B_x = -B_0 \sin \omega t, B_y = 2B_0 \cos \omega t$	(2)	Elliptically polarized wave whose principal axes coincide with the x- and y-axes.
(R)	$B_x = -B_0 \cos \left(\omega t + \frac{\pi}{3} \right), B_y = B_0 \cos \omega t$	(3)	Elliptically polarized wave whose principal axes are inclined at 45° to the x- and y-axes.
(S)	$B_x = -B_0 \cos \left(\omega t + \frac{\pi}{3} \right), B_y = \sqrt{3} B_0 \cos \omega t$	(4)	The major axis makes an angle θ with the x-axis satisfying $\tan 2\theta = \frac{\sqrt{3}}{2}$
		(5)	The axial ratio (ratio of major axis to minor axis) is $\sqrt{\frac{4+\sqrt{7}}{4-\sqrt{7}}}$

(A) $P \rightarrow 1; Q \rightarrow 2; R \rightarrow 3; S \rightarrow 4, 5$

(B) $P \rightarrow 2; Q \rightarrow 1; R \rightarrow 3; S \rightarrow 4, 5$

(C) $P \rightarrow 1; Q \rightarrow 2; R \rightarrow 3, 5; S \rightarrow 4$

(D) $P \rightarrow 3; Q \rightarrow 2; R \rightarrow 1; S \rightarrow 4$

3) A source emits sound of frequency $f_0 = 680$ Hz in still air where the speed of sound is $v = 340$ m/s. In each situation of List-I, determine the wavelength of sound reaching the observer. All velocities are measured relative to the air. Which of the following gives the correct matching ?

	List-I		List-II
(P)	Source moves toward a stationary observer with speed 68 m/s.	(1)	0.375 m
(Q)	Source is stationary, while the observer moves toward the source with speed 34 m/s.	(2)	0.400 m

(R)	Source moves away from a stationary observer with speed 85 m/s.	(3)	0.500 m
(S)	Source moves toward the observer with speed 68 m/s, while the observer simultaneously moves away from the source with speed 34 m/s.	(4)	0.625 m
		(5)	0.750 m

(A) $P \rightarrow 2; Q \rightarrow 3; R \rightarrow 4; S \rightarrow 2$

(B) $P \rightarrow 2; Q \rightarrow 1; R \rightarrow 4; S \rightarrow 3$

(C) $P \rightarrow 3; Q \rightarrow 2; R \rightarrow 5; S \rightarrow 1$

(D) $P \rightarrow 2; Q \rightarrow 3; R \rightarrow 5; S \rightarrow 1$

SECTION-II

1) A metal rod of length 1.50 m is clamped at its midpoint and excited in the third allowed longitudinal mode in a Kundt's tube. The tube contains a gas in which successive dust heaps are initially separated by 2.00 cm. The gas temperature is then increased so that the speed of sound becomes 10% greater, while the rod is simultaneously heated so that its length increases by 0.20%. Assume the longitudinal wave speed in the rod remains unchanged and the rod continues vibrating in the same mode. If the new separation between successive dust heaps is x cm, find $100x$.

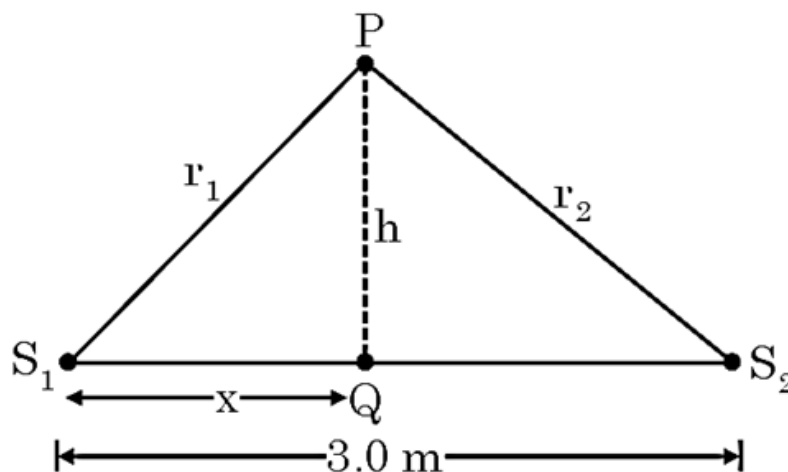
2) A plano-convex lens of radius of curvature $R = 1.20$ m rests on a glass plate, forming Newton's rings in reflected monochromatic light of wavelength 600 nm. A small dust particle trapped between the lens and the plate produces a central film thickness t_0 . The diameter of the dark ring

corresponding to interference order $n = 14$ is 4.80 mm. When a liquid of refractive index $\mu = \frac{4}{3}$ is introduced between the lens and plate, keeping the lens geometry unchanged, the diameter of the dark ring corresponding to the same interference order becomes 3.60 mm. If $t_0 = x \times 10^{-7}$ m, find x .

3) A compound microscope is adjusted for final image at infinity. Its tube length is $L = 18$ cm, eyepiece focal length is $f_e = 6$ cm, and total magnifying power is 90. Take the least distance of distinct vision as $D = 25$ cm. The objective has clear aperture diameter 1.20 cm. The space between specimen and objective is filled with an immersion liquid of refractive index 1.50. Light of wavelength 500 nm is used. Due to an aperture stop, only 80% of the objective radius is effectively

available. Use $\sin \theta = \frac{\alpha_{\text{eff}}}{\sqrt{\alpha_{\text{eff}}^2 + f_0^2}}$, where α_{eff} is the effective aperture. If $d_{\text{min}} = x$ nm, find the value of x .

4) Two coherent point sources S_1 and S_2 , separated by 3.0 m, emit sound of wavelength 1.0 m. The acoustic power of S_2 is four times that of S_1 . At the sources, the oscillation of S_2 leads that of S_1 in phase by $\pi/3$. A point P lies in the plane containing the sources, on one side of the line S_1S_2 . The distances of P from S_1 and S_2 are r_1 and r_2 , respectively. Assume each source behaves as an ideal point source, so that the amplitude of the sound wave varies inversely with distance. At P , complete destructive interference occurs, and $r_1 < 2$ m. Let x be the perpendicular projection distance of P



from S_1 measured along S_1S_2 , as shown :

If $x = \frac{N}{18} \text{ m}$, find the integer N.

5) Two sound waves of frequencies f_1 and f_2 reach a point P. Their individual intensities at P are $9I_0$ and $4I_0$, respectively. At $t = 0$, the resultant intensity at P is $7I_0$ and is increasing. During the next 2.0s, the resultant intensity becomes $19I_0$ exactly 8 times. Neither $t = 0$ nor $t = 2.0\text{s}$ corresponds to

$19I_0$. If the beat frequency is $f_b = \frac{N}{4} \text{ Hz}$, find the integer N.

6) A laser beam of power 15 W is completely incident on a thin flat plate at an angle 37° with the normal to its surface. The plate reflects 72% of the incident energy specularly, absorbs 18%, and transmits the remaining energy without deviation. Calculate the magnitude of the force exerted by the radiation on the plate in the direction normal to its surface. Express your answer in nN, rounded to two decimal places.

PART-3 : CHEMISTRY

SECTION-I (i)

1) Identify the CORRECT statement about sparingly soluble salt at given temperature.

- (A) Solubility increases due to the complex formation of one of the ions with reagent added
- (B) Solubility decreases due to hydrolysis of ion/s of the salt.
- (C) Solubility decreases on dilution.
- (D) Solubility Product constant decreases due to the common ion effect.

2) Choose the incorrect statement of the following:

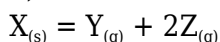
- (A) During corrosion of iron in atmosphere, H^+ ions are produced by dissolving CO_2 of atmosphere in water.
- (B) Alkaline medium promotes the rusting of iron.
- (C) $\text{Pt}|\text{H}_2(\text{P}_1)|\text{H}^+(\text{C}_1)||\text{H}^+(\text{C}_2)|\text{H}_2(\text{P}_1)|\text{Pt}$, cell reaction is spontaneous when $\text{C}_2 > \text{C}_1$ according to the representation.

- (D) Corrosion can be prevented by covering the metal surface with chrome and nickel via electrolysis.

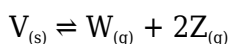
3) Choose the incorrect statement :

- (A) In Leclanche cell $\text{MnO}_2 + \text{NH}_4^+ + \text{e}^- \rightarrow \text{NH}_3 + \text{MnO}(\text{OH})$ takes place at anode
 (B) In mercury cell, the overall reaction does not involve any ion in solution whose concentration can change cell potential during progress of time
 (C) Ni-Cd cell has longer life (more life cycles) than lead storage cell
 (D) $\text{Zn} \rightarrow \text{Zn}^{2+} + 2\text{e}^-$ takes place in Leclanche cell at anode

4) Sufficient amount of a solid X is taken in a rigid vessel at $t^\circ\text{C}$ where it attained the equilibrium.



Total pressure was measured. Now the vessel is evacuated and filled with sufficient amount of another solid V under same conditions where it attained the equilibrium:



Total pressure measured now is found to be double that of previous value. Now, if both X and V solids are allowed to attain their respective equilibrium together in the same vessel at same temperature, select the incorrect statement

- (A) K_p for decomposition reaction of $\text{V}_{(\text{s})} = 8 \times K_p$ for decomposition reaction of $\text{X}_{(\text{s})}$
 In the 3rd case (when both solids are simultaneously establishing their equilibrium),
 (B) $P_Y = \frac{1}{8}P_w$
 (C) P_Y in 3rd case = $\frac{1}{3\sqrt[3]{3}} \times P_Y$ in 1st case
 (D) In the 3rd case, $P_w : P_z = 3 : 8$

SECTION-I (ii)

1) A mixture contains Na_2CO_3 and NaHCO_3 . 2.0 g of this mixture requires 15.0 mL of 1.0 M HCl for neutralization using phenolphthalein indicator. Another 2.0 g sample of the same mixture requires V mL of 1.0M HCl using methyl orange indicator. Which of the following statements is/are CORRECT? (Formula mass of $\text{Na}_2\text{CO}_3 = 106$ g/mol, $\text{NaHCO}_3 = 84$ g/mol)

- (A) Mass of Na_2CO_3 in 2.0 g mixture is 1.59 g
 (B) Mass percentage of NaHCO_3 in the mixture is 20.5%
 (C) The total volume of 1.0M HCl required for complete neutralization using methyl orange is approx. 35.0 mL
 (D) Moles of NaHCO_3 present in 2.0 g mixture is approx. 0.008 mol.

2) The correct statement(s) is/are

- (A) Specific conductance increases while molar conductivity decreases on progressive dilution
 (B) The limiting equivalent conductivity of weak electrolyte cannot be determined exactly by extrapolation of the plot of Λ_{eq} against $\sqrt{c}$

(C) A plot of α^2 versus $\frac{1}{C}$ gives a straight line with slope equals to dissociation constant of weak electrolyte AB ($\alpha \ll 1$)

(D) Kohlrausch law is valid both for strong and weak electrolytes.

3) 2 amperes of current is passed for 16 minutes and 5 seconds through 4 litres of 1M aqueous brine solution using Pt electrodes. Which of the following is/are correct about the process?

(Assume 100% current efficiency)

(A) pH of the solution increases.

(B) The volume of Chlorine gas liberated at anode is 224 ml at 1 atm, 0°C.

(C) The volume of hydrogen gas liberated at cathode is 224 ml at 1 atm, 0°C.

(D) The mass of Na deposited is 0.46 grams.

SECTION-I (iii)

1)

List-I		List-II	
(P)	$\text{PCl}_5(\text{g}) \rightleftharpoons \text{PCl}_3(\text{g}) + \text{Cl}_2(\text{g})$	(1)	$K_p > K_c$ at 300K
(Q)	$2\text{NO}_2(\text{g}) \rightleftharpoons \text{N}_2\text{O}_4(\text{g})$	(2)	$K_p < K_c$ at 300K
(R)	$\text{H}_2(\text{g}) + \text{I}_2(\text{g}) \rightleftharpoons 2\text{HI}(\text{g})$	(3)	Pressure is increased, reaction moves to right side
(S)	$\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightleftharpoons 2\text{NH}_3(\text{g})$	(4)	Inert gas added at constant volume, equilibrium will not be affected
		(5)	Inert gas at constant pressure, equilibrium will shift right

Which of the following represents the correct combination ?

(A) P → 1; Q → 2; R → 2; S → 3

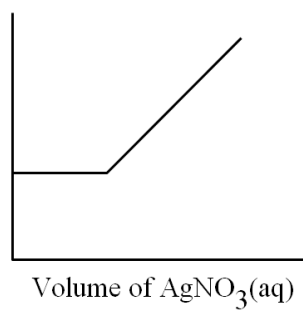
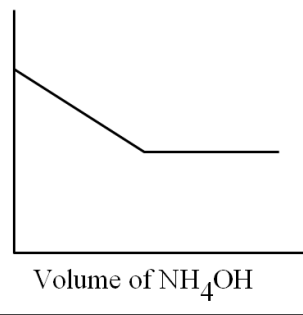
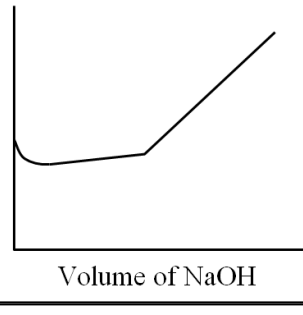
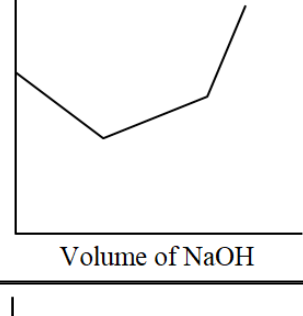
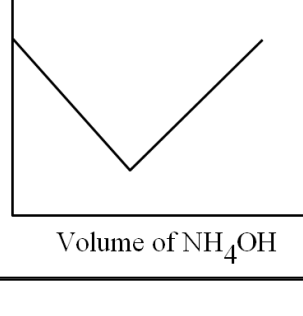
(B) P → 2; Q → 3; R → 4; S → 5

(C) P → 1; Q → 2; R → 4; S → 3

(D) P → 5; Q → 1; R → 2; S → 3

2) Which of the following graphs correctly matched for variation of conductance (C) during titration ?

List-I	List-II
--------	---------

(P)	$\text{AgNO}_3(\text{aq})$ was added to an aq. KCl gradually (assume ionic mobility of NO_3^- & Cl^- is same)	(1)	
(Q)	$\text{HCl}(\text{aq})$ titrated against NH_4OH	(2)	
(R)	$\text{CH}_3\text{COOH}(\text{aq})$ titrated against $\text{NaOH}(\text{aq})$	(3)	
(S)	Titration of a mixture of $[\text{HCl}_{\text{aq}} + \text{CH}_3\text{COOH}_{\text{aq}}]$ against $\text{NaOH}(\text{aq})$	(4)	
		(5)	

(A) $\text{P} \rightarrow 1; \text{Q} \rightarrow 3; \text{R} \rightarrow 5; \text{S} \rightarrow 2$

(B) $\text{P} \rightarrow 5; \text{Q} \rightarrow 2; \text{R} \rightarrow 1; \text{S} \rightarrow 4$

(C) $\text{P} \rightarrow 1; \text{Q} \rightarrow 2; \text{R} \rightarrow 3; \text{S} \rightarrow 4$

(D) $\text{P} \rightarrow 3; \text{Q} \rightarrow 1; \text{R} \rightarrow 4; \text{S} \rightarrow 5$

3) List-I has redox reaction, List-II has equivalent weight of underline reactant.

List-I	List-II
--------	---------

(P)	$[\text{Fe}(\text{CN})_6]^{-4} \rightarrow \text{Fe}^{+3} + \text{CO}_3^{-2} + \text{NO}_3^-$	(1)	$\frac{M}{2}$
(Q)	$\text{Al} + \text{HNO}_3 \rightarrow \text{Al}(\text{NO}_3)_3 + \text{NH}_4\text{NO}_3$	(2)	$\frac{M}{61}$
(R)	$\text{MnO}_2 \rightarrow \text{MnO}_4^- + \text{Mn}^{+2}$	(3)	$\frac{10M}{8}$
(S)	$\text{KMnO}_4 \rightarrow \text{K}_2\text{MnO}_4 + \text{MnO}_2 + \text{O}_2$	(4)	$\frac{8M}{10}$
		(5)	$\frac{5M}{6}$

(A) $P \rightarrow 2; Q \rightarrow 3; R \rightarrow 5; S \rightarrow 4$

(B) $P \rightarrow 2; Q \rightarrow 4; R \rightarrow 5; S \rightarrow 1$

(C) $P \rightarrow 2; Q \rightarrow 3; R \rightarrow 4; S \rightarrow 1$

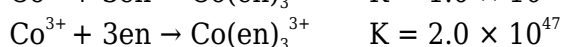
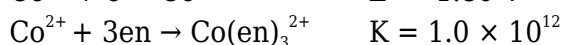
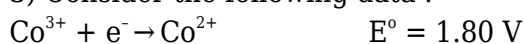
(D) $P \rightarrow 2; Q \rightarrow 3; R \rightarrow 5; S \rightarrow 1$

SECTION-II

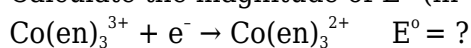
1) The iodine content of a solution was determined by titration with cerium (IV) sulphate in the presence of HCl, in which I^- is converted to ICl . A 200 mL sample of the solution required 15 mL of 0.05 N Ce^{4+} solution, calculate the iodine concentration in original solution in $\frac{\text{g}}{\text{L}}$? [Atomic weight of 'I' is 127 g/mol]

2) An acid-base indicator which is a weak acid has a pK_a value 5.45. At what concentration ratio of CH_3COONa to CH_3COOH would the indicator show a colour halfway between those of its acid and conjugate forms. pK_a of AcOH is 4.75. [$\log 2 = 0.3$; $\log 3 = 0.48$; $\log 5 = 0.7$]

3) Consider the following data :



Calculate the magnitude of E° (in volt) for the half-reaction,

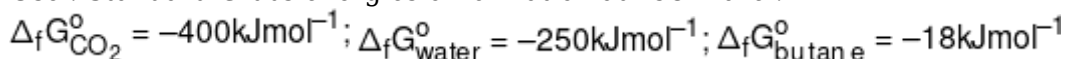


$$\frac{2.303RT}{F}$$

(Given, $\frac{2.303RT}{F} = 0.06$, $\log 2 = 0.3$, $\log 3 = 0.48$, $\log 7 = 0.85$)

4) An electrochemical cell is fueled by the combustion of butane at 1 bar and 298 K. Its cell potential is $\frac{x}{F} \times 10^3$ volts, where F is the Faraday constant. The value of x is _____.

Use : Standard Gibbs energies of formation at 298 K are :



5) The solubility of barium iodate in an aqueous solution prepared by mixing 200 mL of 0.010 M barium nitrate with 100 mL of 0.10 M sodium iodate is $x \times 10^{-6} \text{ mol dm}^{-3}$. The value of 'x' is

Use : Solubility product constant (K_{sp}) of barium iodate = 1.6×10^{-9}

6) Dichloro acetic acid is oxidized to CO_2 , H_2O and Cl_2 by 1.2 equivalents of an oxidizing agent. Same amount of acid can neutralize 'x' moles of NH_3 to give ammonium dichloro acetate. Find 'x'.

ANSWER KEYS

PART-1 : MATHEMATICS

SECTION-I (i)

Q.	1	2	3	4
A.	D	B	C	B

SECTION-I (ii)

Q.	5	6	7
A.	A,C,D	B,C,D	A,B

SECTION-I (iii)

Q.	8	9	10
A.	A	B	C

SECTION-II

Q.	11	12	13	14	15	16
A.	1.00	7.00	64.00	3.00	0.00	20.00

PART-2 : PHYSICS

SECTION-I (i)

Q.	17	18	19	20
A.	A	C	A	B

SECTION-I (ii)

Q.	21	22	23
A.	A,B,C,D	A,B,C	B

SECTION-I (iii)

Q.	24	25	26
A.	A	A	A

SECTION-II

Q.	27	28	29	30	31	32
A.	220.44	18.00	407.37 to 407.43	2.00	8.00	64.60 to 64.80

PART-3 : CHEMISTRY

SECTION-I (i)

Q.	33	34	35	36
A.	A	B	A	D

SECTION-I (ii)

Q.	37	38	39
A.	A,B,C	B,C,D	A,B,C

SECTION-I (iii)

Q.	40	41	42
A.	C	C	D

SECTION-II

Q.	43	44	45	46	47	48
A.	0.23 to 0.25	5.00	0.31 to 0.32	108.92	4.00	0.20

SOLUTIONS

PART-1 : MATHEMATICS

1) Equations $\alpha x^2 + 7x - 2 = 0$ and $2x^2 - 7x - a = 0$ must have atleast one common root.

$$\frac{x^2}{-7\alpha - 14} = \frac{x}{-4 + \alpha^2} = \frac{1}{-7\alpha - 14}$$

$$x = \frac{-7(\alpha + 2)}{\alpha^2 - 4} = \frac{\alpha^2 - 4}{-7(\alpha + 2)}$$

$$49(\alpha + 2)^2 = (\alpha + 2)^2 (\alpha - 2)^2$$

$$\alpha = -2, -5, 9$$

for $\alpha = -2$ both roots are common and for $\alpha = -5, 9$ exactly one root is common.

$$2) (b+c) \left(\left(\frac{a}{(b+c)} \right)^n + 1 \right)^{\frac{1}{n}} + (c+a) \left(\left(\frac{b}{(c+a)} \right)^n + 1 \right)^{\frac{1}{n}} + (a+b) \left(\left(\frac{c}{(a+b)} \right)^n + 1 \right)^{\frac{1}{n}}$$

$$= 2(a + b + c) = 40$$

$$3) \lim_{x \rightarrow 0} \left(2 + \log_{\sec(\frac{x}{2})}^2 \cos \left(\frac{x}{3} \right) \right)^3$$

$$= \lim_{x \rightarrow 0} \left(2 + \left(\frac{\log(\cos \frac{x}{3})}{\log(\sec \frac{x}{2})} \right)^2 \right)^3$$

Now solve

$$\lim_{x \rightarrow 0} \frac{\log(\cos \frac{x}{3})}{\log(\sec \frac{x}{2})} \quad \left(\frac{0}{0} \text{ form} \right)$$

So apply L Hospital rule

$$= \lim_{x \rightarrow 0} \frac{\frac{1}{\cos(x/3)} - \sin(\frac{x}{3}) \cdot \frac{1}{3}}{\frac{1}{\sec(x/2)} \cdot \sec(\frac{x}{2}) \cdot \tan(\frac{x}{2}) \cdot \frac{1}{2}}$$

$$= \lim_{x \rightarrow 0} \frac{-2 \cdot \frac{x}{3} \cdot \sin(\frac{x}{3})}{\frac{(x/2) \cdot (\frac{2}{x})}{(\frac{x}{3}) \cdot \tan(\frac{x}{2}) \cos(\frac{x}{3})}}$$

On taking limit

$$= -\frac{4}{9}$$

Now put in question

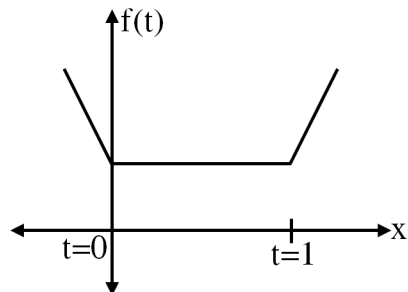
$$= \left(2 + \left(\frac{-4}{9} \right)^2 \right)^3 = \left(\frac{178}{81} \right)^3$$

4)

Given

$$f(t) = \begin{cases} 1-2t, & t < 0 \\ 1, & 0 \leq t \leq 1 \\ 2t-1, & t > 1 \end{cases} \Rightarrow g(x) = \begin{cases} f(x-1), & 0 \leq x \leq 1 \\ 3-x, & 1 < x \leq 2 \end{cases}$$

$$\Rightarrow g(x) = \begin{cases} 3-2x, & 0 \leq x \leq 1 \\ 3-x, & 1 < x \leq 2 \end{cases}$$



□ The function $g(x)$ is discontinuous at $x = 1$, as

$\lim_{x \rightarrow 1} g(x)$ does not exist

So, $g(x)$ is non-derivable at $x = 1$.

$$\ell n(L) = \lim_{n \rightarrow \infty} \frac{1}{n} \ell n \left(\frac{f\left(a + \frac{1}{n}\right)}{f(a)} \right)$$

5)

$$= \lim_{n \rightarrow \infty} \frac{1}{n^2} \left[\frac{\ell n f\left(a + \frac{1}{n}\right) - \ell n f(a)}{\frac{1}{n}} \right]$$

= 0 [□ $n(f(x))$ is differentiable]

(C) Take log and apply L-hospital Rule.

$$6) \lim_{x \rightarrow 0} \left(\frac{\sin(\sin x) - \sin x}{ax^5 + bx^3 + c} \right) = -\frac{1}{12}$$

$c = 0$

$$\lim_{x \rightarrow 0} \left(\frac{2 \sin\left(\frac{\sin x - x}{2}\right) - \cos\left(\frac{\sin x + x}{2}\right)}{\left(\frac{\sin x - x}{2}\right)} \right) \left(\frac{\sin x - x}{2(ax^5 + bx^3)} \right)$$

$$\lim_{x \rightarrow 0} \left(\frac{2 \sin\left(\frac{\sin x - x}{2}\right) - \cos\left(\frac{\sin x + x}{2}\right)}{\left(\frac{\sin x - x}{2}\right)} \right) \left(\frac{\sin x - x}{x^3} \right) \left(\frac{1}{ax^2 + b} \right)$$

$$= \frac{-1}{6b} = \frac{-1}{12} \Rightarrow b = 2$$

$a \in \mathbb{R}, b = 2, c = 0$

7)

We have $f(2) = 0$, so $h(2) = f(2)g(2) = 0$. The derivative is given by:

$$h'(2) = \lim_{x \rightarrow 2} \frac{f(x)g(x)}{x-2} = \lim_{x \rightarrow 2} \left[(x-2) \cos\left(\frac{\pi x}{4}\right) + |x-2| \right] g(x)$$

Evaluating the leading factor:

$$\lim_{x \rightarrow 2} \left[(x-2) \cos \left(\frac{\pi x}{4} \right) + |x-2| \right] = 0 + 0 = 0$$

• **Statements (A) and (B):** If $g(x)$ is bounded in an open interval around $x = 2$ (which also holds whenever $\lim_{x \rightarrow 2} g(x)$ exists), then by the Squeeze Theorem the product of an infinitesimal function with a bounded function converges to 0:

$$h'(2) = 0$$

Both **(A)** and **(B)** are **TRUE**.

• **Statements (C) and (D):** Let $g(x) = 1$ for all $x \in \mathbb{R}$. Then g is bounded, $h'(2) = 0$, but $\lim_{x \rightarrow 2} g(x) = 1 \neq 0$, disproving (D). Furthermore, setting $g(x) = 1$ for $x \neq 2$ and $g(2) = 5$ preserves $h'(2) = 0$ while making g discontinuous at $x = 2$, disproving (C).

Correct Options: (A), (B)

$$\begin{aligned} 8) (P) \quad & \lim_{x \rightarrow \infty} \frac{1}{\pi} \tan^{-1}(x^2 - x^4) \\ &= \frac{\tan^{-1}(-\infty)}{\pi} = \frac{-\frac{\pi}{2}}{\pi} = -\frac{1}{2} \end{aligned}$$

$$(Q) \quad \lim_{x \rightarrow \infty} \frac{e^{x \ln 2}}{e^x} = \lim_{x \rightarrow \infty} \frac{2^x}{e^x} = \lim_{x \rightarrow \infty} \left(\frac{2}{e} \right)^x = 0$$

$$\begin{aligned} (R) \quad & y = fff(x) \\ & y' = f'(f(f(x))) \cdot f'(f(x)) \cdot f'(x) \\ & y'(0) = f'(f(f(0))) \cdot f'(f(0)) \cdot f'(0) \\ & = 2 \cdot 2 \cdot 2 = 8 \end{aligned}$$

$$(S) \quad \lim_{x \rightarrow 2^-} \frac{[x]}{\{x\}} = \lim_{x \rightarrow 2^-} \frac{1}{1} = 1$$

9)

(P) Continuity at $x = 0$

$$\lim_{x \rightarrow 0^+} A x^a \left(x - \frac{1}{x} \right) = 1$$

$$A + a = -1 + 1 = 0$$

(Q) for $a = 1$

$$k = \lim_{x \rightarrow 0^-} \frac{\ln(1+x) - \ln(1-bx)}{x} = 1 + b$$

$$\lim_{x \rightarrow 0^+} \frac{\sqrt{1+2x} - \sqrt{1-2x}}{\sin x} = 2$$

$$k - b = 1$$

$$(R) \quad c = \lim_{x \rightarrow 0} \frac{1 - \cos 2x}{x^2} = 2$$

$$(S) \quad \lim_{x \rightarrow 0} \frac{\sqrt{1 + \sqrt{1 + kx^4}} - \sqrt{2}}{x^4} = \frac{k}{4\sqrt{2}}$$

$$k = \frac{1}{2} \Rightarrow 6k = 3$$

10)

(P) At $x = 1$, continuity

$$a + b = 1$$

$$2a = \left(\frac{1}{x}\right)_{x=1} = -1 \Rightarrow a = -\frac{1}{2}, b = \frac{3}{2}$$

$$(Q) f(0) = \lim_{x \rightarrow 0} x^{p-1} \sin \frac{1}{x}$$

$$p = 2$$

$$(R) f(x) = \max \{|x|, x^2, x^3\}$$

$$x = -1, 0, 1$$

(S) Differentiability at 0

$$a = 0$$

$$1 = b \Rightarrow b = 1$$

11) $|x + y + z| = |x| + |y| + |z|$, if they all have same sign or the terms are equal to zero.

$$\square |1 - \log_{1/6}^x| + |-\log_2^x| + 2 = |1 - \log_{1/6}^x + (-\log_2^x) + 2|$$

$$1 - \log_{1/6}^x \geq 0 \text{ and } -\log_2^x \geq 0$$

$$\square 1 - \log_{1/6}^x \geq 0 \Rightarrow \log_{1/6}^x \leq 1$$

$$\Rightarrow x \geq \left(\frac{1}{6}\right)^1 \Rightarrow x \geq 1/6$$

$$\square -\log_2^x \geq 0 \Rightarrow \log_2^x \leq 0$$

$$\Rightarrow x \leq (2)^0 \Rightarrow x \leq 1$$

$$\square x \in \left[\frac{1}{6}, 1\right]$$

$$12) \sqrt{\left[x + \left[\frac{x}{2}\right]\right]} + \left[\sqrt{\{x\}} + \left[\frac{x}{3}\right]\right] = 3$$

$\square$ when $\left[\frac{x}{3}\right]$ is an integer then definitely $\sqrt{\left[x\right] + \left[\frac{x}{2}\right]}$ is also an integer.

$$\text{So } \sqrt{\left[x\right] + \left[\frac{x}{2}\right]} = 2 \text{ and } \left[\frac{x}{3}\right] = 1 \text{ (and check like this)}$$

$$\left[x\right] + \left[\frac{x}{2}\right] = 4, \left[\frac{x}{3}\right] = 1 \Rightarrow x \in [3, 6)$$

When $x \in [3, 4)$

$$\left[x\right] = 3, \left[\frac{x}{2}\right] = 1$$

So, $x \in [3, 4)$ satisfies.

When $x \in [4, 5) = 4 \left[\frac{x}{2}\right] = 2 \Rightarrow \left[x\right] + \left[\frac{x}{2}\right] = 6 \neq 4$ not satisfies, similarly on checking all possibilities we have only $x \in [3, 4)$.

$$\square a = 3, b = 4$$

$$13) f(x) = \frac{1 - \sin x}{(\pi - 2x)^2} \times \frac{\log \sin x}{\log(1 + (\pi - 2x)^2)}$$

$$\lim_{h \rightarrow 0} \frac{1 - \cosh}{4h^2} \times \frac{\log(\cosh)}{\log(1 + 4h^2)}$$

$$\lim_{h \rightarrow 0} \frac{1 - \cosh}{4h^2} \times \frac{\log(1 + (\cosh - 1))}{\log(1 + 4h^2)} \times \frac{(\cosh - 1)}{(\cosh - 1)}$$

$$\frac{1}{16} \lim_{h \rightarrow 0} \frac{1 - \cosh}{h^2} \times \frac{\log(1 + (\cosh - 1))}{(\cosh - 1)} \times \frac{(\cosh - 1)}{h^2} \times \frac{4h^2}{\log(1 + 4h^2)}$$

$$\frac{1}{16} \times \left(\frac{1}{2}\right) \times 1 \times \left(-\frac{1}{2}\right) \times 1 = -\frac{1}{64}$$

14) Taking logarithm of both sides, we get $\log y = x \left[\log \left(1 + \frac{1}{x} \right) \right]$

$$\Rightarrow \frac{1}{y} y_1(x) = \frac{x^2}{x+1} \left(-\frac{1}{x^2} \right) + \log \left(1 + \frac{1}{x} \right)$$

$$= -\frac{1}{x+1} + \log \left(1 + \frac{1}{x} \right) \quad (i)$$

Since $y(2) = \left(1 + \frac{1}{2} \right)^2 = \frac{9}{4}$

so $y_1(2) = \left(\frac{9}{4} \right) \left(-\frac{1}{3} + \log \frac{3}{2} \right)$

Multiplying (i) by y and then differentiating, we get

$$y_2(x) = y_1(x) \left(-\frac{1}{x+1} + \log \left(1 + \frac{1}{x} \right) \right)$$

$$+ y(x) \times \left(\frac{1}{(x+1)^2} + \frac{x}{x+1} \left(-\frac{1}{x^2} \right) \right)$$

So $y_2(2) = y_1(2) \left(-\frac{1}{3} + \log \frac{3}{2} \right) + y(2) \left(\frac{1}{9} - \frac{1}{6} \right)$

$$= \left(\frac{9}{4} \right) \left(-\frac{1}{3} + \log \frac{3}{2} \right)^2 - \frac{1}{8}$$

15) $\ell n y = \left(\frac{1}{2} \ell n (1 + 2x) + \frac{1}{4} \ell n (1 + 4x) + \dots \right) - \left[\frac{1}{3} \ell n (1 + 3x) + \frac{1}{5} \ell n (1 + 5x) + \dots \right] \quad \dots(1)$

Note : This is equivalent to taking the natural logarithm of

$$y = \frac{(1 + 2x)^{1/2} (1 + 4x)^{1/4} \dots}{(1 + 3x)^{1/3} (1 + 5x)^{1/5} \dots}$$

Differentiating both sides with respect to x using the chain rule :

$$\frac{y'}{y} = \frac{1}{2} \left(\frac{1}{1 + 2x} \right) (2) + \frac{1}{4} \left(\frac{1}{1 + 4x} \right) (4) + \dots - \left[\frac{1}{3} \left(\frac{1}{1 + 3x} \right) (3) + \frac{1}{5} \left(\frac{1}{1 + 5x} \right) (5) + \dots \right] = 0 \quad \dots(2)$$

16)

Let common roots are x_1 and x_2 then by equations

$$x_1 + x_2 + \alpha = 5$$

$$x_1 + x_2 + \beta = 2$$

$$\text{or } \alpha - \beta = 3$$

again by given equations

$$3x^2 - 3x + r - q = 0 \begin{cases} x_1 \\ x_2 \end{cases}$$

$$x_1 + x_2 = 1$$

$$\text{so } \alpha = 4, \beta = 2$$

$$\text{Now by limit } a = 16, b = 4$$

$$a + b = 20$$

PART-2 : PHYSICS

17)

Correct Answer is (A)

18)

Correct Answer is (C)

19)

Correct Answer is (A)

20)

Correct Answer is (B)

21)

Correct Answer is (A,B,C,D)

22)

Correct Answer is (A,B,C)

23)

Correct Answer is (B)

24)

Correct Answer is (A)

25)

Correct Answer is (A)

26)

Correct Answer is (A)

27)

Correct Answer is (220.44)

28)

Correct Answer is (18.00)

29)

Correct Answer is (407.37 to 407.43)

30)

Correct Answer is (2.00)

31)

Correct Answer is (8.00)

32)

Correct Answer is (64.60 to 64.80)

PART-3 : CHEMISTRY

33) (A) Correct: Complexation removes free metal ions, driving the dissolution equilibrium forward and increasing solubility. (B) Incorrect: Hydrolysis of an ion removes it from solution, shifting equilibrium forward to increase solubility.

(C) Incorrect: Dilution generally increases the total amount of salt dissolved.

(D) Incorrect: K_{sp} is a thermodynamic equilibrium constant dependent only on temperature, not on the common ion effect.

34) (A) Correct: Dissolved CO_2 forms H_2CO_3 , generating H^+ ions which promote the reduction reaction in rust formation.

(B) Incorrect: Alkaline medium deplete H^+ ions, inhibiting corrosion.

(C) Correct: For a concentration cell $\text{Pt}/\text{H}_2(\text{P}_1)/\text{H}^+(\text{C}_1)//\text{H}^+(\text{C}_2)/\text{H}_2(\text{P}_1)/\text{Pt}$, $E_{\text{cell}} = 0.059 \log \left(\frac{C_2}{C_1} \right)$.
If $C_2 > C_1$, $E_{\text{cell}} > 0$ (spontaneous).

(D) Correct: Corrosion can be prevented by electroplating of chrome or nickel.

35) (A) Incorrect: Zinc oxidation ($\text{Zn} \rightarrow \text{Zn}^{2+} + 2\text{e}^-$) occurs at the anode, while $\text{MnO}_2 + \text{NH}_4^+ +$

$e^- \rightarrow \text{MnO(OH)} + \text{NH}_3$ occurs at the cathode.

(B) Correct: The overall reaction in a mercury cell does not contain ions whose concentrations change, keeping its cell potential constant.

(C) Correct: Ni-Cd secondary batteries have a significantly longer cycle life than lead-acid batteries.

(D) Correct: Oxidation of Zn to Zn^{2+} occurs at the anode in a Leclanché cell.

36) First experiment : $X_{(s)} \rightleftharpoons Y_{(g)} + 2Z_{(g)}$

$$\bullet P_Y = p_1, P_Z = 2p_1 \Rightarrow P_{\text{total},1} = 3p_1$$

$$\bullet K_{p1} = (p_1)(2p_1)^2 = 4p_1^3$$

Second experiment : $V_{(s)} \rightleftharpoons W_{(g)} + 2Z_{(g)}$

$$\bullet P_{\text{total},2} = 2 \times P_{\text{total},1} = 6p_1 \Rightarrow 3p_2 = 6p_1 \Rightarrow p_2 = 2p_1$$

$$\bullet K_{p2} = (p_2)(2p_2)^2 = 4p_2^3 = 4(2p_1)^3 = 32p_1^3$$

$$\frac{K_{p2}}{K_{p1}} = \frac{32p_1^3}{4p_1^3} = 8$$

$$\bullet \frac{K_{p2}}{K_{p1}} = 8 \Rightarrow K_{p2} = 8K_{p1} \text{ (A is correct)}$$

Simultaneous equilibrium:

$$\bullet P_Y = p_x, P_W = p_v, P_Z = 2p_x + 2p_v$$

$$\bullet K_{p1} = p_x(2p_x + 2p_v)^2 = 4p_x(p_x + p_v)^2 = 4p_1^3$$

$$\bullet K_{p2} = p_v(2p_x + 2p_v)^2 = 4p_v(p_x + p_v)^2 = 32p_1^3$$

$$\bullet \text{Dividing the two equations : } \frac{p_x}{p_v} = \frac{1}{8} \Rightarrow P_Y = \frac{1}{8}P_W \text{ (B is correct)}$$

$$\bullet \text{Since } p_v = 8p_x, \text{ substituting gives : } 4p_x(9p_x)^2 = 324p_x^3 = 4p_1^3 \Rightarrow p_x^3 =$$

$$\frac{p_1^3}{81} \Rightarrow P_x = \frac{p_1}{\sqrt[3]{81}} = \frac{p_1}{3\sqrt[3]{3}}$$

$$\bullet P_w : P_z = p_v : 2(p_x + p_v) = 8p_x : 2(9p_x) = 8 : 18 = 4 : 9 \text{ (D is incorrect)}$$

37) Phenolphthalein indicator : Na_2CO_3 is neutralized upto NaHCO_3 only. ($n_{\text{HCl}} = \frac{1}{2}n_{\text{Na}_2\text{CO}_3}$).

Moles of HCl = $1.0 \times 0.015 \text{ L} = 0.015 \text{ mol}$

$n_{\text{Na}_2\text{CO}_3} = 0.015 \text{ mol}$

Mass of $\text{Na}_2\text{CO}_3 = 0.015 \times 106 = 1.59 \text{ gm}$

methyl orange indicator : Neutralized both Na_2CO_3 fully and NaHCO_3

Total $V_{\text{HCl}} = 2 \times V_{\text{phenolphthalein}} + V_{\text{NaHCO}_3} = 34.88 \text{ ml}$

38) (A) Incorrect: Progressive dilution decreases specific conductance (κ) due to fewer ions per unit volume, but increases molar conductivity.

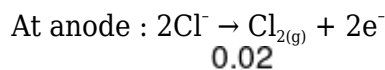
(B) Correct: Extrapolation of Λ_{eq} vs $\sqrt{C}$ does not give limiting molar conductivity for weak electrolytes because the curve steepens asymptotically near zero concentration.

(C) Incorrect: According to Ostwald's Dilution Law ($\alpha = \sqrt{K_a/C}$), $\alpha^2 = K_a/C$. Thus plotting α^2 vs $1/C$ yields a straight line through the origin with slope equal to K_a .

(D) Correct: Kohlrausch's law of independent migration of ions applies to both strong and weak electrolytes.

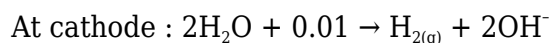
39) · Charge passed (Q): $I \times t = 2 \text{ A} \times (16 \times 60 + 5)\text{s} = 2 \times 965 \text{ s} = 1930 \text{ C}$

· Moles of electrons : $\frac{1930}{96500} = 0.02 \text{ mol}$



Moles of $\text{Cl}_2 = \frac{0.02}{2} = 0.01 \text{ mol}$

Volume at STP = $0.01 \times 22400 \text{ mL} = 224 \text{ mL}$

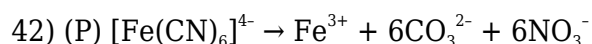


OH^- of $\text{H}_2 = 0.01 \text{ mol} \Rightarrow$ Volume at STP = 224 mL

OH^- ions generated $\Rightarrow$ pH increases.

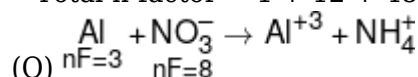
40) Theoretical

41) Theoretical

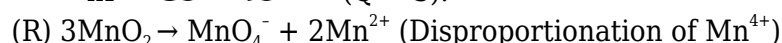


• Oxidation state changes: $\text{Fe}^{2+} \rightarrow \text{Fe}^{3+}$ ($\Delta = 1$), $6\text{C}^{2+} \rightarrow 6\text{C}^{4+}$ ($\Delta = 12$), $6\text{N}^{3-} \rightarrow 6\text{N}^{5+}$ ($\Delta = 48$)

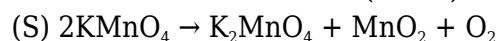
• Total n-factor = $1 + 12 + 48 = 61 \Rightarrow E = \frac{M}{61}$ (P $\rightarrow$ 2)



$nF = \frac{24}{30} = \frac{8}{10}$ (Q $\rightarrow$ 3).



• n-factor = $\frac{6}{5} \Rightarrow E = \frac{5M}{6}$ (R $\rightarrow$ 5)



• n-factor calculation gives $E = \frac{M}{2}$ (S $\rightarrow$ 1)

43) Reaction: $\text{I}^- \rightarrow \text{ICl}$ (Oxidation state of Iodine goes from -1 to +1, n-factor = 2).

meq of $\text{Ce}^{4+} = 15 \text{ mL} \times 0.05 \text{ N} = 0.75 \text{ meq}$

meq of $\text{I}^- = 0.75 \Rightarrow$ Moles of $\text{I}^- = \frac{0.75 \times 10^{-3}}{2} = 3.75 \times 10^{-4} \text{ mol}$

Mass of Iodine in 200 mL (0.2 L) = $3.75 \times 10^{-4} \times 127 \text{ g} = 0.047625 \text{ g}$

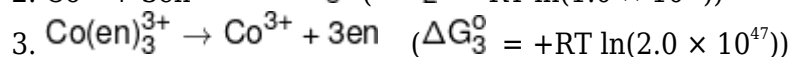
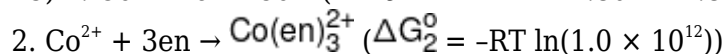
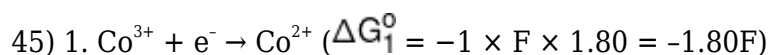
Concentration in g/L = $\frac{0.047625 \text{ g}}{0.2 \text{ L}} = 0.238 \text{ g/L}$

44) Halfway color point: $[\text{HIn}] = [\text{In}^-] \Rightarrow \text{pH} = \text{pK}_{\text{In}} = 5.45$.

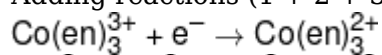
Buffer equation: $\text{pH} = \text{pK}_a + \log \left(\frac{[\text{CH}_3\text{COONa}]}{[\text{CH}_3\text{COOH}]} \right)$

$5.45 = 4.75 + \log(R) \Rightarrow \log(R) = 0.70$

Since $\log 5 = 0.7$, the required ratio $R = 5$.



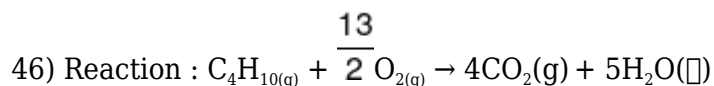
Adding reactions (1 + 2 + 3):



$$\Delta G^\circ = \Delta G_1^\circ + \Delta G_2^\circ + \Delta G_3^\circ$$

$$-FE^\circ = -1.80F + 2.303RT \log \left(\frac{2 \times 10^{47}}{1 \times 10^{12}} \right)$$

$$E^\circ = -0.318 \text{ V}$$



Number of transferred electrons $n = 26$.

$$\Delta_r G^\circ = 4\Delta_f G^\circ(\text{CO}_2) + 5\Delta_f G^\circ(\text{H}_2\text{O}) - \Delta_f G^\circ(\text{C}_4\text{H}_{10})$$

$$\Delta_r G^\circ = 4(-400) + 5(-250) - (-18) = -1600 - 1250 + 18 = -2832 \text{ kJ/mol} = -2832 \times 10^3 \text{ J/mol}$$

$$\Delta_r G^\circ = -nF E_{\text{cell}}^\circ \Rightarrow E_{\text{cell}}^\circ = \frac{2832 \times 10^3}{26F} = \frac{108.92 \times 10^3}{F} \text{ V}$$

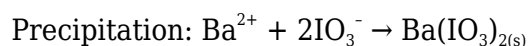
Therefore, $x = 108.92$

47) Initial moles :

- $\text{Ba}^{+2} : 0.200 \text{ L} \times 0.010 \text{ M} = 0.002 \text{ mol}$

- $\text{IO}_3^- : 0.100 \text{ L} \times 0.10 \text{ M} = 0.010 \text{ mol}$

- Total Volume = 300 mL - 0.3 L



- Ba^{2+} completely precipitates (limiting reagent).

- Remaining $\text{IO}_3^- = 0.010 - 2(0.002) = 0.006 \text{ mol}$

$$\text{Concentrated } [\text{IO}_3^-] = \frac{0.006}{0.3} = 0.02$$

$$K_{\text{sp}} = [\text{Ba}^{2+}][\text{IO}_3^-]^2 = 1.6 \times 10^{-9} = s \times (0.02)^2 = s \times (4 \times 10^{-4})$$

$$s = \frac{1.6 \times 10^{-9}}{4 \times 10^{-4}} = 4 \times 10^{-6} \text{ M}$$

$$\text{Comparing with } x \times 10^{-6}, x = 4$$

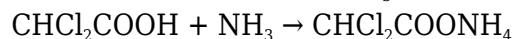
48) Oxidation reaction of dichloroacetic acid (CHCl_2COOH):



Redox n-factor for oxidation of 1 mole of CHCl_2COOH is 6.

$$\text{Equivalents} = \text{moles} \times \text{n-factor} \Rightarrow 1.2 = \text{moles} \times 6 \Rightarrow \text{moles of acid} = 0.2 \text{ mol.}$$

Acid-base reaction with NH_3 :



$$\text{Ratio is } 1 : 1 \Rightarrow x = \text{moles of } \text{NH}_3 = \text{moles of acid} = 0.2$$